

Introduction to Machine Learning / Assignment – 1**Abdulmalik Hussein 445801958**

Suppose we have the following dataset with one response (target) variable y and two predictor variables X_1 and X_2 :

X_1	Y
60	140
62	155
67	159
70	179
71	192
72	200
75	212
78	215

Using the Normal Equations for Simple Linear regression, calculate the model's parameters (w and b).

$X_1 (x)$	$Y (y)$	x^2	$x \cdot y$
60	140	3600	8400
62	155	3844	9610
67	159	4489	10653
70	179	4900	12530
71	192	5041	13632
72	200	5184	14400
75	212	5625	15900
78	215	6084	16770

$$\begin{aligned}
 \text{Sum of } x (\sum x) &= 555 \\
 \text{Sum of } y (\sum y) &= 1.452 \\
 \text{Sum of } x^2 (\sum x^2) &= 38,767 \\
 \text{Sum of } xy (\sum xy) &= 101,895
 \end{aligned}
 \quad \left\{ \begin{array}{l} n=8 \\ \text{data points} \end{array} \right.$$

$$X^T X = \begin{bmatrix} n & \sum x \\ \sum x & \sum x^2 \end{bmatrix} = \begin{bmatrix} 8 & 555 \\ 555 & 38,767 \end{bmatrix}$$

$$X^T Y = \begin{bmatrix} \sum y \\ \sum xy \end{bmatrix} = \begin{bmatrix} 1.452 \\ 101,895 \end{bmatrix}$$

$$\begin{aligned}
 \det(X^T X) &= n \sum x^2 - (\sum x)^2 \\
 &= (8)(38,767) - (555)^2 = 310,136 - 308,025 \\
 &= 2,111
 \end{aligned}$$

Slope (w):

$$\begin{aligned}
 w &= \frac{n \sum xy - \sum x \sum y}{\det(X^T X)} = \frac{815,160 - 805,860}{2111} \\
 &= \frac{9300}{2111} \approx 4.4055
 \end{aligned}$$

Intercept (b):

$$\begin{aligned}
 b &= \frac{\sum x^2 \sum y - \sum x \sum xy}{\det(X^T X)} \\
 b &= \frac{56289684 - 56551725}{2111} = \frac{-262041}{2111}
 \end{aligned}$$

$$\approx -124.1312$$

Final parameters:

$$\text{Slope (w)}: 9300/2111 \approx 4.4055$$

$$\text{Intercept (b)}: \approx -124.1312$$

Fitted Model Equation:

$$\hat{y} = 4.4055 x_1 - 124.1312$$